

Devops

1. DevOps Foundations and Operating Systems

- **YouTube:**

- Content Engaging, highly practical deep-dive into basic-to-advanced Linux commands.

- TechWorld with Nana - DevOps Roadmap – The gold standard introductory overview of the DevOps lifecycle.

- **Hands-on Labs:**

- SadServers – Real Linux servers where you must troubleshoot broken configurations.

- OverTheWire (Bandit) – Command-line wargame designed to build absolute terminal mastery.

- **Documentation:**

- The Linux Documentation Project – The foundational architecture manual for Linux environments.

2. Version Control, Git, and CI/CD Pipelines

- **YouTube:**

- Git & GitHub Crash Course by Traversy Media – Complete end-to-end breakdown of local and remote version control.

- GitHub Actions Tutorial by Kunal Kushwaha – High-yield, clear implementation guide for modern automated pipelines.

- **Hands-on Labs:**

- Learn Git Branching – An interactive visual sandbox simulation for branching and merging strategies.

- KodeKloud CI/CD Labs – In-browser operational testing sandboxes for GitHub Actions, Jenkins, and GitLab.

- **Documentation:**

- Pro Git Book Official Docs – The comprehensive authority covering advanced workflows and internals.

3. Cloud Platforms, IaC, and Provisioning

- **YouTube:**

- freeCodeCamp AWS Certified Cloud Practitioner – Thorough architectural dive into core enterprise cloud design.

- [HashiCorp Terraform Associate Course by Andrew Brown](#) – Direct, comprehensive implementation course for Infrastructure as Code.

- **Hands-on Labs:**

- [AWS Workshop Studio](#) – Step-by-step architectural templates created by internal AWS product teams.

- [Terraform Tutorials by HashiCorp Developer](#) – Self-paced tracks with native sandbox sandboxes.

- **Documentation:**

- [AWS Architecture Center](#) & [Terraform Registry Documentation](#) – Essential guides for configuring infrastructure state files securely.

4. Configuration Management and Secret Handling

- **YouTube:**

- [Ansible 101 by Jeff Geerling](#) – The ultimate step-by-step playbook composition series created by a core community expert.

- [HashiCorp Vault Tutorial by Bryan Krausen](#) – Production-grade secrets management, engine setup, and integration practices.

- **Hands-on Labs:**

- [Killercoda Ansible Scenarios](#) – Live Linux terminals configured with multi-node setups to test operational playbooks instantly.

- [HashiCorp Vault Interactive Tracks](#) – Guided setup labs targeting token emission, rotation policies, and app integration.

- **Documentation:**

- [Ansible User Guide](#) & [HashiCorp Vault Production Hardening Guide](#) – Strict compliance manuals detailing least-privilege paradigms.

5. Monitoring, Observability, and Log Management

- **YouTube:**

- [Prometheus & Grafana Setup by TechWorld with Nana](#) – Clear architectural explanation of pull-based metrics vs dashboards.

- [The ELK Stack Crash Course by Official Elastic](#) – Thorough walk-through on parsing raw logs into structured indexes.

- **Hands-on Labs:**

- [Grafana Play](#) – Live dashboard exploration platform demonstrating query builders and alerting profiles.

- [Elastic Cloud Trial Sandbox](#) – Cloud environment populated with dummy application payloads to practice custom log ingestion workflows.

- **Documentation:**

- [Google SRE Book](#) – The foundation of observability philosophy, outlining modern SLA, SLO, and SLI management frameworks.

6. Artifact Management, Delivery, and Containers

- **YouTube:**

- [Docker Tutorial for Beginners by Programming with Mosh](#) – Pristine explanation of layering, image micro-sizing, and volume configurations.
- [Kubernetes Tutorial for Beginners \(TechWorld with Nana\)](#) – Accessible 4-hour blueprint detailing stateful sets, pods, and deployments.

- **Hands-on Labs:**

- [Play with Docker](#) & [Play with Kubernetes](#) – Multi-node interactive cluster runtimes operating completely inside your browser.
- [Katacoda / Killercoda Kubernetes Interactive Scenarios](#) – Direct configuration drills designed for debugging real-time multi-container applications.

- **Documentation:**

- [Kubernetes Tasks Docs](#) – The prescriptive implementation blueprint and reference handbook for operational engineers.

7. Service Mesh, Traffic Control, and Resilience

- **YouTube:**

- [Istio Service Mesh Explained by Rawkode Academy](#) – Complete structural walk-through detailing sidecar proxies, mutual TLS, and mesh control planes.
- [Hussein Nasser Network Engineering Channel](#) – Deep dive into proxies, reverse proxies, layer 4/7 load balancers, and reverse engineering networks.

- **Hands-on Labs:**

- [Istio Getting Started Scenarios](#) – Sandbox guides modeling canary rollouts and traffic shaping profiles.
- [Envoy Proxy Sandboxes](#) – Docker-compose scripts modeling real-world failure patterns and rate-limiting loops.

- **Documentation:**

- [Istio Traffic Management Architecture](#) – Foundational patterns detailing network resilience, circuit-breaking parameters, and routing structures.

8. Incident Response, Reliability, and SRE

- **YouTube:**

- [SREcon Presentation Archive \(USENIX\)](#) – Direct production retro studies from tech giants dealing with cascading system failures at scale.
- [PagerDuty Engineering Outage Postmortems](#) – Professional walk-throughs covering active responder rotation schedules and coordination setups.

- **Hands-on Labs:**

- **Chaos Mesh Interactive Playground** – Inject synthetic network latency and pod failures directly into active clusters to evaluate cluster health.
- **Gremlin Free Certification Tracks** – Sandbox drills centered around blast-radius modeling and game-day blast containment strategies.

- **Documentation:**

- Google Site Reliability Engineering Handbook – The definitive operational standard defining incident response command structures.

9. Platform Excellence, Automation, and Developer Experience (IDPs)

- **YouTube:**

- Internal Developer Platforms (IDP) Guide by Backstage Team – Core structural walkthrough of service catalogs, plugin management, and developer portals.
- Platform Engineering vs DevOps by Internalized Industry Panels – Clarifying how to construct tailored golden paths that abstract cloud complexity.

- **Hands-on Labs:**

- Spotify Backstage Demo App Playground or Demo app Visual interaction workspace for evaluating software templates and registration components.
- **Humanitec Platform Orchestrator Paths** – Blueprint drills modeling configuration management and infrastructure templates.

- **Documentation:**

- **PlatformEngineering.org Whitepapers** – The foundational architecture manual mapping enterprise golden paths, drivers, and platform topologies.